

Raunak Kumar

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Education

Cornell University

PHD IN COMPUTER SCIENCE

Ithaca, New York

2018 - Present

University of British Columbia

B.Sc. (WITH DISTINCTION) IN COMPUTER SCIENCE

Vancouver, British Columbia

2013 - 2018

- Advisor: Mark Schmidt.
- A+ graduating average.

Awards

- 2020 **NSERC Alexander Graham Bell Canada Graduate Scholarships-Doctoral (CGS D)** CA \$105,000 (declined)
- 2020 **NSERC Postgraduate Scholarships-Doctoral (PGS D)** CA \$63,000
- 2019 **Cornell Computer Science Teaching Assistant Award**
- 2018 **Cornell University Fellowship** US \$30,400
- 2018 **NSERC Canada Graduate Scholarships-Master's** CA \$17,500 (declined)
- 2018 **Rick Sample Memorial Research Scholarship**
- 2018 **CRA Outstanding Undergraduate Researcher Award** Honorable Mention
- 2017 **UBC Computer Science Teaching Assistant Award**
- 2017 **NSERC Undergraduate Student Research Award** CA \$4,000
- 2013-2017 **UBC Science Scholar / Dean's Honors List**
- 2014 **J. Fred Muir Memorial Scholarship in Science** CA \$200
- 2014 **Trek Excellence Scholarship** CA \$1,500
- 2013 **UBC Chancellor's Scholar Award**

Experience

Department of Computer Science, Cornell

GRADUATE RESEARCH ASSISTANT

Ithaca, USA

September 2020 - Present

Working with Prof. Robert D. Kleinberg broadly on machine learning and theoretical computer science.

Microsoft

DATA SCIENTIST INTERN

Bellevue, USA

May 2020 - August 2020

Worked in the Microsoft Search, Assistant and Intelligence (MSAI) team that powers Microsoft Office 365 on improving topic suggestions for users. Developed multiple unsupervised and supervised learning algorithms that obtained significant improvements in precision and recall. Also developed new evaluation metrics and visualizations that provide a more detailed look into the performance of an algorithm on this task.

Department of Computer Science, Cornell

GRADUATE RESEARCH ASSISTANT

Ithaca, USA

February 2019 - December 2019

Worked with Prof. Austin R. Benson on developing neural network models for temporal graphs and hypergraphs. Also developed fast deterministic and randomized sampling algorithms for weighted subgraph retrieval in large graphs.

Department of Computer Science, UBC

UNDERGRADUATE RESEARCH ASSISTANT

Vancouver, Canada

May 2017 - May 2018

Worked with Prof. Mark Schmidt on optimization for machine learning. Proved the first non-asymptotic convergence rate of expectation-maximization (EM) and developed improved variants of the EM algorithm. Also collaborated on projects in computer vision and consulted for a local company on detecting dangerous gases based on sensor measurements as part of a smart air purification system.

Google

SOFTWARE ENGINEERING INTERN

Mountain View, USA

May 2016 - August 2016

Worked on a distributed datastore trigger utility in C++. This utility provides an extensible way to maintain triggers, which are used to schedule and perform asynchronous actions. It was used by an internal team to remove 1 billion triggers from storage.

Publications

Conference Proceedings

- [1] Frederik Kunstner, Raunak Kumar, and Mark Schmidt. "Homeomorphic-Invariance of EM: Non-Asymptotic Convergence in KL Divergence for Exponential Families via Mirror Descent". In: *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*. **Oral**. 2021.
- [2] Raunak Kumar*, Paul Liu*, Moses Charikar, and Austin R. Benson. "Retrieving Top Weighted Triangles in Graphs". In: *Proceedings of the 13th ACM International Conference on Web Search and Data Mining (WSDM)*. 2020.

Workshop Papers

- [3] Raunak Kumar and Mark Schmidt. "Convergence Rate of Expectation-Maximization". The 10th Neural Information Processing Systems (**NeurIPS**) Workshop on Optimization for Machine Learning (**OPT**). 2017.

Teaching

Department of Computer Science

Cornell University

TEACHING ASSISTANT

January 2019 - May 2019

Worked as a TA for Mathematical Foundations for The Information Age taught by Prof. John Hopcroft. My responsibilities included handling course administration, leading and coordinating the team of TAs, holding office hours and grading. I was awarded the Cornell Computer Science Teaching Assistant Award in recognition of my accomplishments and contributions in this role.

Computer Science 490: Problem Solving Seminar

University of British Columbia

INSTRUCTOR

Jan 2017 - April 2017

As an undergrad, co-taught a full 3 credit student seminar focusing on practical applications of advanced algorithms and data structures. My responsibilities included lecturing, writing lecture notes and creating assignments. Website:

<https://www.students.cs.ubc.ca/~cs-490/2016W2/>

Department of Computer Science

University of British Columbia

TEACHING ASSISTANT

September 2014 - April 2018

Worked as a TA for many courses, including Machine Learning (graduate and undergraduate), Advanced Algorithm Design and Analysis, Intermediate Algorithm Design and Analysis, Advanced Operating Systems, Introduction to Computer Systems, and Models of Computation. My responsibilities included lecturing, grading, and holding office hours and tutorials. I was awarded the UBC Computer Science Teaching Assistant Award in recognition of my accomplishments and contributions in this role.

Honors

PROGRAMMING CONTESTS

- 2017 **Coach** ACM-ICPC PacNW Regionals (Div. 2) - placed 1st and 3rd (2017)
- 2015-2017 **Contestant** ACM-ICPC PacNW Regionals - placed 13th (2016) and 15th (2017) in Div. 1, and 6th in Div. 2 (2015)
- 2016-2017 **Contestant** Microsoft College Code Competition - placed 6th (2016) and 2nd (2017)
- 2016-2017 **Contestant** NAIPC (Open Division USA/Canada) - placed 27th (2016) and 15th (2017)
- 2017 **Contestant** SFU Winter Programming Contest - placed 8th
- 2015 **Contestant** Calgary Collegiate Programming Competition (Div. 2) - placed 3rd

Service and Outreach

SERVICE

- 2020 **Reviewer** COLT

OUTREACH

- 2019 **Girls' Adventures in Math** Volunteered for an outreach event for upper elementary and middle school girls. *Ithaca, USA*
- 2019 **Expand Your Horizons** Taught Scratch to girls in 7th - 9th grades to get them interested in STEM fields. *Ithaca, USA*

Extracurricular Activity

Computer Science Graduate Organization

Cornell University

PRESIDENT

October 2020 - Present

Leading the representative body for computer science graduate students. Working on mechanisms to help integrate new students into the department in light of the pandemic, supporting underrepresented minority applicants applying for graduate school, and improving the TA experience for graduate students.

Computer Science Graduate Organization

Cornell University

SPECIAL REPRESENTATIVE FOR WASTE MANAGEMENT

May 2019 - September 2020

Worked to develop a culture of reuse and composting in the department, and reduce the amount of landfill waste generated, especially single-use plastic from food events. Engaged with the Cornell Sustainability Office to initiate campus wide compost dropoff spots and organize an educational sustainability themed seminar for graduate students in the department.